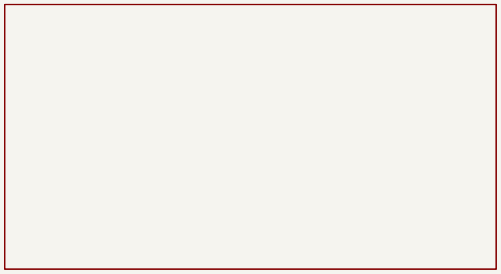
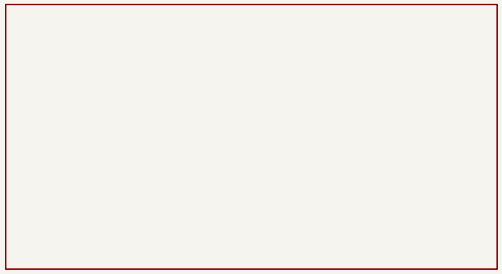
PowerMCU



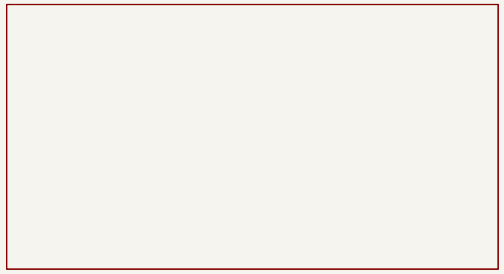
檔案： PowerMCU.kicad_sch

Measurement



檔案： Measurement.kicad_sch

DisplayUI



檔案： DisplayUI.kicad_sch

Marc USB integrated measurement board — D0 / NOT FOR FABRICATION

USB -> LOLIN 3V3 -> ADG706 + ADS1115 + Adafruit 4814 FeatherWing (3V3 only)

DSSC -> 16 nominal load candidates -> ADG706 -> GND; ADS1115 senses DUT voltage

Global net labels connect sheets. All parts instantiated; actual pins shown on child sheets.

No harvesting, battery, relay, audio or PUF. Current firmware remains unchanged.

Gate A incomplete: DUT Voc/Isc, MUX official specification, procurement and contacts.

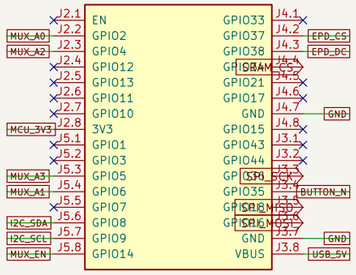
PCB is mechanical baseline + interfaces + off-board staging; NO SIGNAL ROUTING.

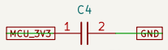
Based on Marc Dusseiller IVcurve_tester f9a75be; new integrated design

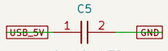
Sheet: /
File: Marc_USB.kicad_sch

Title: Marc USB — Integrated draft

Size: A4	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 1/4









USB ONLY: use module USB-C; no second carrier USB supply and no external 3V3 input.

VBUS and MCU_3V3 are module outputs. EPD 3V3_OUT must NEVER join MCU_3V3.

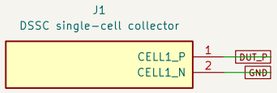
Onboard ME6211C33: current/thermal budget and recessed USB plug fit NOT qualified.

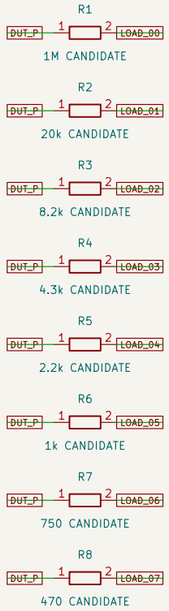
J2/J5/J4/J3 pin names match WEMOS V1.0.0 schematic. Antenna / height review pending.

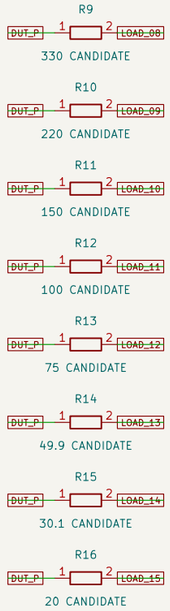
Sheet: /PowerMCU/
File: PowerMCU.kicad_sch

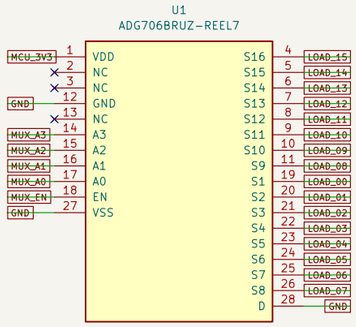
Title: Marc USB – PowerMCU – DRAFT NOT FOR FAB

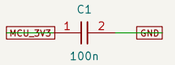
Size: A3	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 2/4

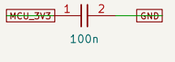




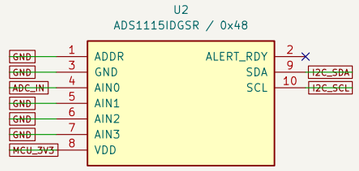


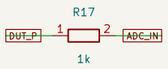


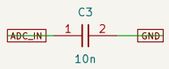


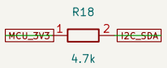














DRAFT: ADG706 pin map cross-checked with upstream symbol/PCB; official datasheet review BLOCKED.

DUT is NOT a power source. No overvoltage or reverse-input qualification. Voc/Isc REQUIRED.

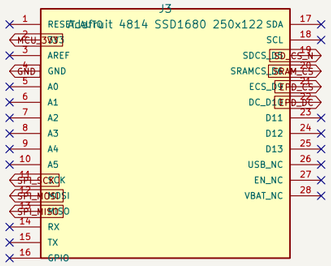
R1..R16 are independent nominal candidates, NOT recovered Marc physical BOM; new calibration required.

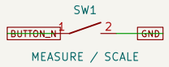
ADS1115: ADDR=GND => 0x48. GAIN_ONE +/-4.096V is NOT a 4.096V-safe input at VDD=3.3V.

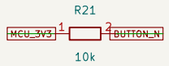
MUX_EN on GPIO14, 10k pulldown: boot disabled. Firmware patch required. No TPI MOSFET.

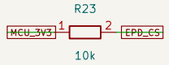
R17/C3 ADC filter only: NOT MUX input protection. Disconnect DUT before removing USB power.

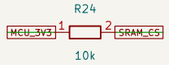
Sheet: /Measurement/		
File: Measurement.kicad_sch		
Title: Marc USB – Measurement – DRAFT NOT FOR FAB		
Size: A3	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 3/4











Adafruit 4814: SSD1680 / 250x122 / red–black–white; full Feather 16+12 mating pins.

3.3V ONLY at pin2. USB/VBAT/EN unused. Factory SJ1..SJ4 closed; SDCS pulled high; no SD.

Onboard auto-reset; RESET header pin1 NC. BUSY testpad not wired. MCU GPIO10/GPIO7 NC.

SW1 short press = scan; long press = scale menu; serial/web scale configuration patch required.

Collector A/B polarity, zebra material, force/resistance and plunger stack remain UNQUALIFIED.

Sheet: /DisplayUI/
File: DisplayUI.kicad_sch

Title: Marc USB – DisplayUI – DRAFT NOT FOR FAB

Size: A3	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 4/4